

Note: If a number is divisible by p as well as q , where p and q are co-primes, then the given number is divisible by pq .

If p and q are not co-primes, then the given number need not be divisible by pq , even when it is divisible by both p and q .

Ex. 36 is divisible by both 4 and 6, but it is not divisible by $(4 \times 6) = 24$, since 4 and 6 are not co-primes.

VI. Factorial of a Number

Let n be a positive integer.

Then, the continued product of first n natural numbers is called factorial n , denoted by $n!$ or $\underline{n}$.

Thus, $n! = n(n-1)(n-2) \dots \dots \dots 3.2.1$

Ex. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Note: $0! = 1$

VII. Modulus of a Number

$$|x| = \begin{cases} x, & \text{when } x \geq 0 \\ -x, & \text{when } x < 0 \end{cases}$$

Ex. $|-5| = 5$, $|4| = 4$, $|-1| = 1$, etc.

VIII. Greatest Integral Value

The greatest integral value of an integer x , denoted by $[x]$, is defined as the greatest integer not exceeding x .

Ex. $[1.35] = 1$, $\left[\frac{11}{4}\right] = \left[2\frac{3}{4}\right] = 2$, etc.

IX. Multiplication BY Short cut Methods

1. Multiplication By Distributive Law:

(i) $a \times (b + c) = a \times b + a \times c$ (ii) $a \times (b - c) = a \times b - a \times c$

Ex. (i) $567958 \times 99999 = 567958 \times (100000 - 1) = 567958 \times 100000 - 567958 \times 1$
 $= (56795800000) - 567958 = 56795232042$.

(ii) $978 \times 184 + 978 \times 816 = 978 \times (184 + 816) = 978 \times 1000 = 978000$.

2. Multiplication of a Number By 5^n :

Put n zeros to the right of the multiplicand and divide the number so formed by 2^n .

Ex. $975436 \times 625 = 975436 \times 5^4 = \frac{9754360000}{16} = 609647500$.

X. Division Algorithm or Euclidean Algorithm

If we divide a given number by another number, then:

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

Important Facts:

1. (i) $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .
- (ii) $(x^n - a^n)$ is divisible by $(x + a)$ for all even values of n .
- (iii) $(x^n + a^n)$ is divisible by $(x + a)$ for all odd values of n .

2. To find the highest power of a prime number p in $n!$

Highest power of p in $n! = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \dots + \left[\frac{n}{p^r}\right]$, where $p^r \leq n < p^{r+1}$

SOLVED EXAMPLES

Ex. 1. Simplify: (i) $8888 + 888 + 88 + 8$

(ii) $715632 - 631104 - 9874 - 999$

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Sol. (i) 8888 (ii) Given $\text{exp} = 715632 - (631104 + 9874 + 99)$
 888 $= 715632 - 641077 = 74555$.

$$\begin{array}{r} 8888 \\ 888 \\ 88 \\ + 8 \\ \hline 9872 \end{array}$$

$$\begin{array}{r} 631104 \\ 9874 \\ + 99 \\ \hline 641077 \end{array}$$

$$\begin{array}{r} 715632 \\ - 641077 \\ \hline 74555 \end{array}$$